TRANSMISSION LINES & WAVEGUIDES (ECA15)

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EXP NO. 3

Designing a Quarter-Wave Transformer by Varying Frequency and Input Impedance

AIM

To design a quarter-wave transformer for input impedance Zin = 50 Ω and load impedance ZL = 100 Ω, and analyze performance as frequency and input impedance are varied.

APPARATUS / SOFTWARE REQUIRED

SCILAB Version 6.1.1

PROCEDURE

• Open SCILAB software.

• Open File → New.

• Enter the input values.

• Type and execute the program.

• Record the displayed results.

CALCULATION

Transformer characteristic impedance: Zt = √(Zin × ZL) Quarter-

wave length: L = λ/4 = c/(4f), where c = 3 × 10■ m/s.

TEST CASE 1: VARYING FREQUENCY

Zin = 50 Ω and ZL = 100 Ω. Frequency values: 100, 200, 300 and 500 MHz.

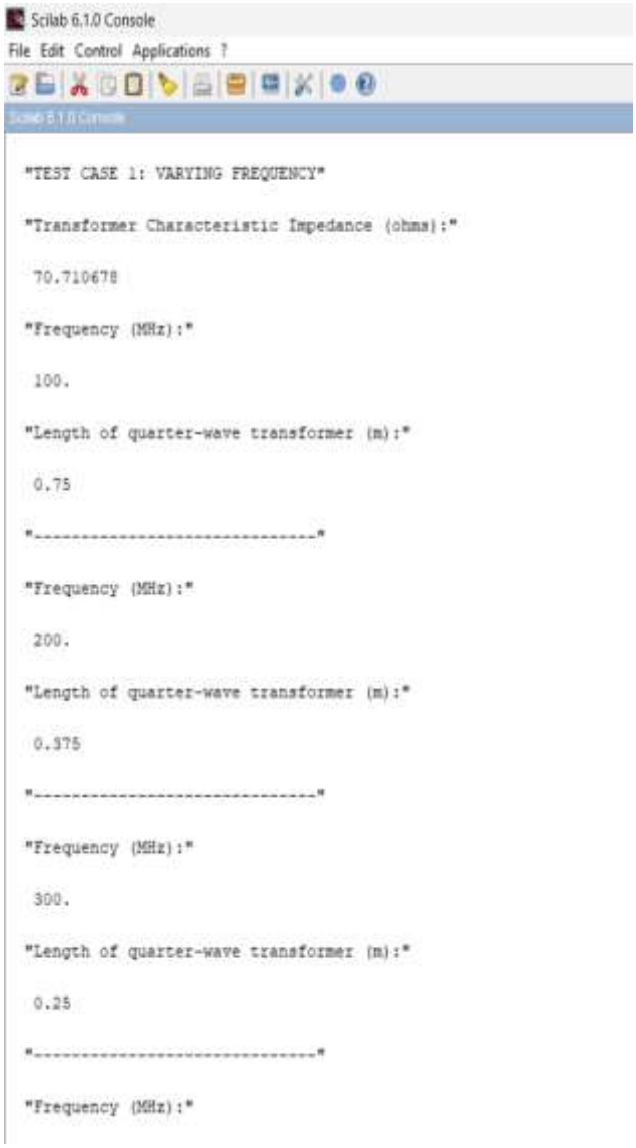
CODE

```
clc;
clear;
Zin = 50;
ZL = 100;
f_values = [100 200 300 500];
Zt = sqrt(Zin * ZL);

disp("TEST CASE 1: VARYING FREQUENCY");
disp("Transformer Characteristic Impedance
(ohms):");
disp(Zt);
```

```
for k = 1:4
f = f_values(k);
L = 300/(4*f);
disp("Frequency (MHz):");
disp(f);
disp("Length of quarter-wave transformer
(m):");
disp(L);
disp("------------------------------");
end
```

OUTPUT SCREENSHOT





Zin (Ω)	ZL (Ω)	Zt (Ω)
50	100	70.7107

RELATED TABLE 1: CHARACTERISTIC IMPEDANCE

RELATED TABLE 2: FREQUENCY VARIATION

Frequency (MHz)	Quarter-wave length (m)
100	0.7500
200	0.3750
300	0.2500
500	0.1500

RESULT – TEST CASE 1

The transformer characteristic impedance is 70.7107 Ω. Quarter-wave length decreases as frequency increases.

TEST CASE 2: VARYING INPUT IMPEDANCE

At fixed frequency 2 GHz and load impedance ZL = 100 Ω, input impedance varies through 25, 50, 75 and 100 Ω.

CODE

```
clc;

clear;

// Fixed parameters

f = 2000;  // Frequency in MHz

ZL = 100;

Zin_values = [25 50 75 100];

disp("TEST CASE 2: VARYING INPUT
IMPEDANCE");

for k = 1:4

   Zin = Zin_values(k);

   Zt = sqrt(Zin * ZL);

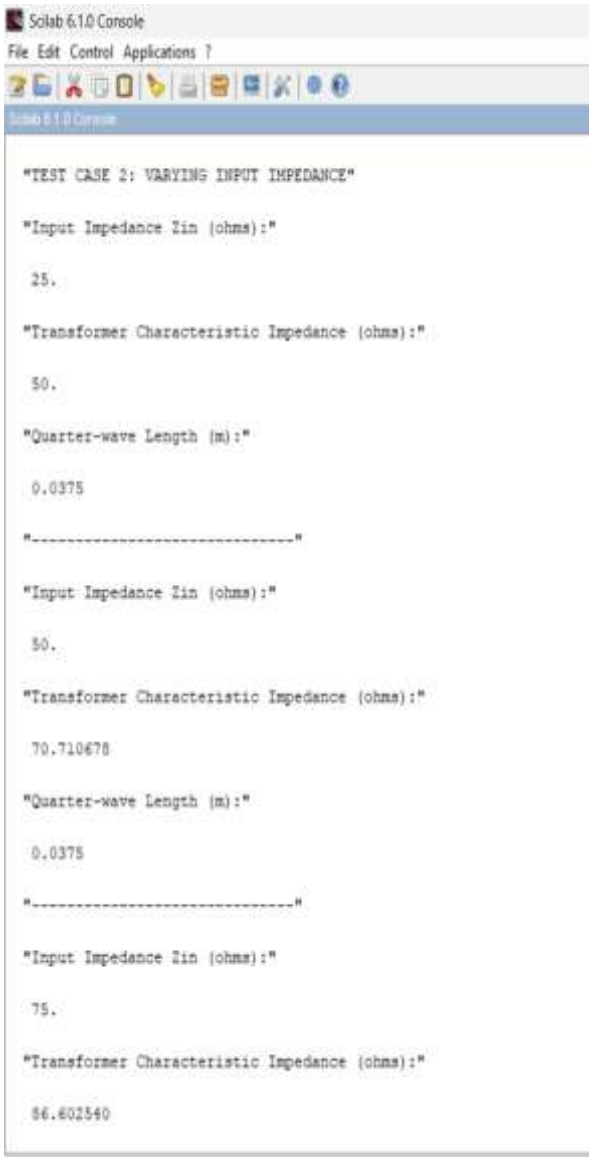
   L = 300/(4*f);

   disp("Input Impedance Zin (ohms):");

   disp(Zin);
```

```
    disp("Transformer Characteristic Impedance
(ohms):");
    disp(Zt);
    disp("Quarter-wave Length (m):");
    disp(L);
    disp("------------------------------");
end
```

OUTPUT SCREENSHOT :





TABLE

Zin (Ω)	ZL (Ω)	Zt (Ω)	Length (m)
50	100	70.7107	0.0375
75	100	86.6025	0.0375
100	100	100.0000	0.0375

RESULT – TEST CASE 2

The transformer characteristic impedance increases with input impedance. At fixed 2 GHz, the quarter-wave length remains 0.0375 m.

OVERALL RESULT

A quarter-wave transformer was analyzed using SCILAB. The quarter-wave length decreased with increasing frequency in Test Case 1. In Test Case 2, characteristic impedance varied with input impedance, while the quarter-wave length remained constant at the fixed operating frequency.